Topic 2.3 - Exponential Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ______________ Class: ______________ Date: ______________

Why this topic matters

An exponential function is controlled by a starting scale, a multiplicative base, and possible transformations. A complete analysis names the horizontal asymptote, specifies whether the graph approaches it from above or below, and distinguishes growth or decay from increasing or decreasing behavior after reflections.

Learning targets

- Analyze the structure, domain, range, monotonicity, concavity, and end behavior of exponential functions.
- Connect equal-interval proportional change to the parameters of an exponential formula.
- Describe and construct transformed exponential functions using precise asymptotic language.

Language and structure

Term	Meaning in this topic
base	The positive constant $b \neq 1$ in ab^x that controls proportional change.
horizontal asymptote	A horizontal line approached by the graph as x moves toward an end of the domain.
monotonicity	Whether a function is increasing or decreasing over an interval.
concavity	Whether slopes are increasing or decreasing; exponential graphs do not have turning points.
transformation	A shift, reflection, stretch, or compression applied to a parent function.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Two functions have the same horizontal asymptote $y = 6$. One passes through $(0, 9)$ and rises; the other passes through $(0, 3)$ and falls as x increases. Sketch a plausible shape for each and explain which parameter signs are forced.

Concept 1: The parent family $f(x) = ab^x$

Core idea

For $a \neq 0$, $b > 0$, and $b \neq 1$, the domain is all real numbers and the horizontal asymptote is $y = 0$.

- If $a > 0$ and $b > 1$, the function is positive, increasing, and concave up.
- If $a > 0$ and $0 < b < 1$, the function is positive, decreasing, and concave up.
- A negative a reflects the graph across the x-axis, reversing the sign and monotonicity.

Worked example - annotate the reasoning

Problem. Analyze $f(x) = 5(1/2)^x$: state domain, range, intercept, monotonicity, concavity, asymptote, and end behavior.
Model reasoning. Domain: all real numbers. Range: $(0, \infty)$. y-intercept $(0, 5)$. The function is decreasing and concave up. Horizontal asymptote: $y = 0$. As $x \to \infty$, $f(x) \to 0^+$; as $x \to -\infty$, $f(x) \to \infty$.

Check your understanding

Independent

Analyze $g(x) = -3(4)^x$ without graphing.

Concept 2: Transformations and asymptotes

Core idea

In $f(x) = a\,b^{x-h} + k$, the horizontal asymptote is $y = k$ and the anchor point is $(h, a + k)$.

- h shifts the graph horizontally; k shifts it vertically.
- The sign of a determines whether values lie above or below the asymptote.
- The range is (k, ∞) when $a > 0$ and $(-\infty, k)$ when $a < 0$.

Worked example - annotate the reasoning

Problem. Describe the graph of $f(x) = -2(3)^{x-1} + 4$, including the asymptote, range, intercepts, and end behavior.
Model reasoning. Asymptote $y = 4$; range $(-\infty, 4)$. Anchor point $(1, 2)$. y-intercept is $-2/3 + 4 = 10/3$. The x-intercept satisfies $3^{x-1} = 2$, so $x = 1 + \log_3 2$. The graph decreases, approaches 4 from below as $x \to -\infty$, and tends to $-\infty$ as $x \to \infty$.

Check your understanding

Independent

For $p(x) = 7(0.4)^{x+2} - 5$, state the asymptote, anchor point, range, and monotonicity.

Concept 3: Constructing an exponential function from features

Core idea

When a horizontal asymptote is known, subtract it before using ratios.

- If $f(x) = ab^x + k$, then $f(x) - k = ab^x$.
- Two points with known k determine a and b when the adjusted outputs are nonzero and have a consistent ratio.
- Check that the resulting base is positive and not equal to 1.

Worked example - annotate the reasoning

Problem. An exponential function has horizontal asymptote $y = 2$ and passes through $(0, 8)$ and $(2, 56)$. Find the function.
Model reasoning. Use $f(x) = ab^x + 2$. From $f(0) = 8$, $a = 6$. From $f(2) = 56$, $6b^2 = 54$, so $b = 3$. Therefore $f(x) = 6(3)^x + 2$.

Check your understanding

Independent

Find $f(x) = ab^x - 4$ if $f(1) = 2$ and $f(3) = 50$.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

State the range and horizontal asymptote of $f(x) = -7(0.2)^{x-3} + 9$.

Exit ticket 2

No calculator

Find the anchor point of $g(x) = 4(5)^{x+1} - 2$.

Exit ticket 3

No calculator

Why can an exponential graph not cross its horizontal asymptote in the form $ab^{x-h} + k$ with $a \neq 0$?